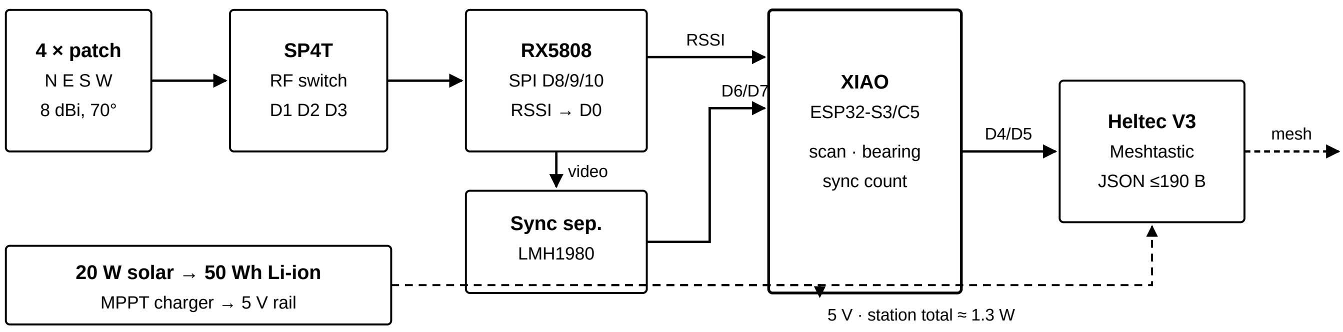
DRONE-MESH-5PLUS · LEVEL 1 STATION · BENCH GUIDE

Level 1 Station v2

Sector direction finding and video fingerprinting on the RX5808 analog FPV receiver. Bench build and verification guide.

Document	docs/LEVEL1-V2-HARDWARE.md	**Firmware**	`level1-analog-fpv/`
Branch	`level1-station`	**Date**	2026-09-09
Status	**Firmware implemented — hardware not yet bench validated**	**Station**	________________

SIGNAL CHAIN AT A GLANCE



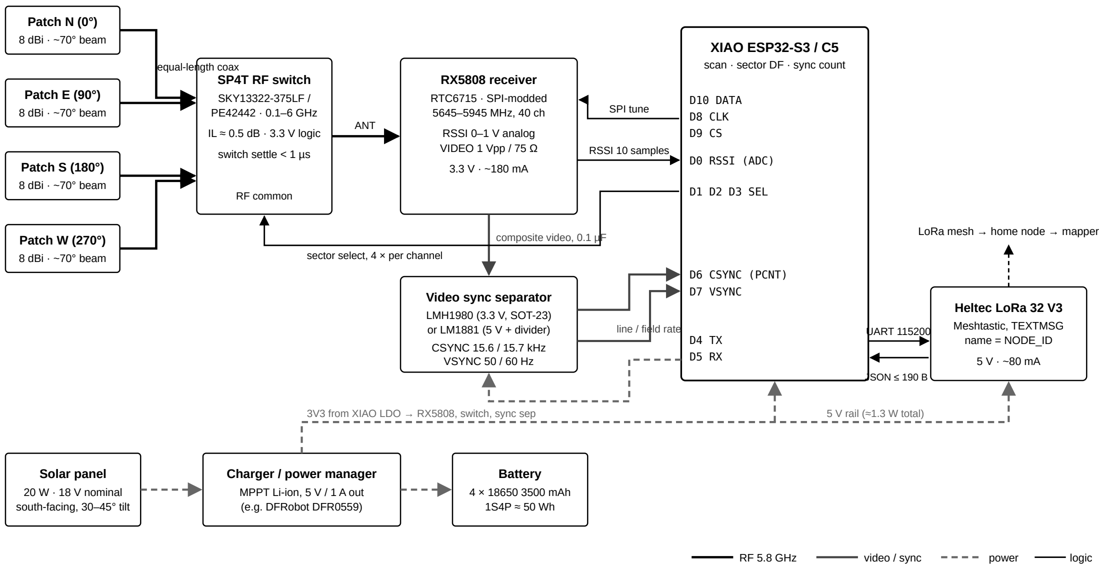
Simplified chain. The full wiring diagram with pin numbers, part numbers and the power tree is on sheet 2.

PARTS, AT A GLANCE

QTY	ITEM	KEY SPEC / PART	HAVE	≈ COST
1	RX5808 receiver	SPI-modded, 5.8 GHz	☐	$8
4	Patch antenna	5.8 GHz, 8 dBi, ~70°, SMA	☐	$32
1	SP4T RF switch	SKY13322-375LF / PE42442	☐	$3
5	u.FL–SMA pigtail	**equal length**, ≤ 15 cm	☐	$10
1	Sync separator	LMH1980 (3.3 V) or LM1881	☐	$3
1	XIAO MCU	ESP32-S3 or ESP32-C5	☐	$9
1	Heltec LoRa 32 V3	Meshtastic, TEXTMSG 115200	☐	$20
1	Solar panel	20 W, 18 V nominal	☐	$25
1	Charger / power manager	MPPT, 5 V / 1 A out	☐	$15
4	18650 Li-ion	3500 mAh, 1S4P ≈ 50 Wh	☐	$24
1	IP65 enclosure + brackets	≈200×150×100 mm, 10–15° wedge	☐	$28
—	Passives	0.1 µF, 10 µF, 100 nF ×4, 2 kΩ ×3, 1000 µF	☐	$3

> **Before you start:** three values in this guide are unverified and must be measured on the bench, not trusted — the RSSI millivolt-to-dBm calibration points, the SP4T control truth table, and the XIAO ESP32-C5 pin map (the C5 column on sheet 2 comes from the Arduino variant and contradicts earlier firmware). Full BOM on sheet 5, procedure on sheet 6.

1 · Full block diagram



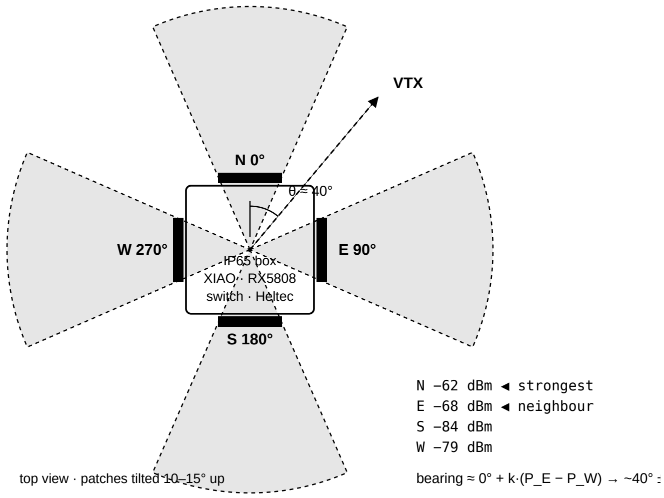
Four patches feed an SP4T switch whose common port replaces the RX5808 whip. The XIAO tunes the receiver over SPI, reads RSSI once per sector, and counts the sync separator's line and field pulses. The XIAO's 3V3 regulator feeds the receiver, switch and sync separator; the 5 V rail feeds the XIAO and Heltec.

WIRING LIST

FROM	TO	XIAO PIN	S3 GPIO	C5 GPIO	NOTES	✓
RX5808 DATA	XIAO	`D10`	`9`	`10`	bit-bang SPI	☐
RX5808 CLK	XIAO	`D8`	`7`	`8`	bit-bang SPI	☐
RX5808 CS	XIAO	`D9`	`8`	`9`	active LOW	☐
RX5808 RSSI	XIAO	`D0`	`1`	`1`	ADC, 0–1 V	☐
Switch V1	XIAO	`D1`	`2`	`0`	2 kΩ series	☐
Switch V2	XIAO	`D2`	`3`	`25`	2 kΩ series	☐
Switch V3	XIAO	`D3`	`4`	`7`	spare if 2-line decode	☐
RX5808 VIDEO	Sync sep. IN	—	—	—	0.1 µF series, 75 Ω term	☐
Sync sep. CSYNC	XIAO	`D6`	`43`	`11`	edge-count interrupt	☐
Sync sep. VSYNC	XIAO	`D7`	`44`	`12`	edge-count interrupt	☐
XIAO TX	Heltec RX	`D4`	`5`	`23`	UART 115200	☐
XIAO RX	Heltec TX	`D5`	`6`	`24`	UART 115200	☐
Patches N/E/S/W	Switch RF1–4	—	—	—	**equal-length coax**	☐
Switch common	RX5808 ANT	—	—	—	shortest run	☐
5 V rail	XIAO 5V, Heltec 5V	—	—	—	1000 µF at Heltec	☐
XIAO 3V3	RX5808, switch, sync sep.	—	—	—	10 µF + 100 nF each	☐
GND	all	—	—	—	single star point	☐

> **GPIO numbers are from the Arduino core's XIAO variants**, which the firmware uses through `config.h`. Wire by the D-labels on the silkscreen. **C5:** earlier firmware assumed D0 = GPIO2, D4 = GPIO6, D5 = GPIO7, which disagrees with the variant column above; the variant is trusted (its D6/D7 are the C5's UART0 defaults, like the S3's). Continuity-test D0, D4 and D5 on the first C5 board and write the result here: ______________________

2 · Sector layout and bearing



Top view. Each patch covers about 70°, so an emitter always sits inside one sector and near the edge of a neighbour.

METHOD

Amplitude comparison. Take the strongest sector `k` and its two neighbours; with powers in dB the bearing is

```
bearing = az[k] + K · (P[k+1] − P[k−1])
```

`K`, in degrees per dB, comes from the measured patch pattern. A bench sweep of a VTX around the box every 15° fills the lookup table.

Report `bearing_deg` and `bearing_sigma_deg`; sigma grows when the two neighbours are nearly equal, or the peak is weak.

PER-CHANNEL SCAN LOOP

```
tune(channel); delay(TUNE_SETTLE_MS)
for sector in N,E,S,W:
    select(sector)
    delayMicroseconds(200)
    stats[sector] = read_rssi_stats()
hit = max(stats).mean_raw >= THRESHOLD
```

Sweep time 40 × (30 + 4 × ~2 ms) ≈ 1.9 s, unchanged in practice. Switching settles in under a microsecond, far inside the 30 ms PLL settle already paid per channel.

VIDEO FINGERPRINT

On the peak channel only, after the sweep: hold the tuner, count CSYNC edges with the PCNT peripheral for 100 ms and time VSYNC edges for 200 ms.

LINE RATE	FIELD RATE	VERDICT	MEANING
`15 734 ± 30 Hz`	`59.94 Hz`	**`NTSC`**	real analog video carrier
`15 625 ± 30 Hz`	`50 Hz`	**`PAL`**	real analog video carrier
no stable train	—	**none**	carrier without video — Wi-Fi, ISM junk, or a bare beacon

`sync_q` (0–100) is the fraction of 10 ms windows whose count landed within ±2 of the expected value: a quality figure that does not depend on the RSSI curve at all. The fingerprint `fp` is `<video>/<line_hz>/<freq_peak>`, e.g. `NTSC/15736/5742`, where the peak frequency comes from stepping the synthesizer ±10 MHz in 2 MHz steps around the hit. Camera crystals and VTX offsets are stable per aircraft over a flight, so two stations reporting the same fingerprint are watching the same drone even when their peak channels differ.

3 · Power budget and output

LOAD	CURRENT	POWER	NOTES
XIAO ESP32-S3/C5, radios off, scanning	~45 mA @ 3.3 V	0.25 W	via the 5 V rail and on-board LDO
RX5808	~180 mA @ 3.3 V	0.60 W	dominant load; cannot be duty-cycled without losing detections
SP4T switch + LMH1980	< 10 mA	0.03 W	the entire DF and fingerprint add-on
Heltec V3, Meshtastic, mostly RX	~80 mA @ 5 V	0.40 W	+ ~120 mA during LoRa TX bursts
Station total		**≈ 1.3 W**	**≈ 31 Wh/day.** 20 W panel at 3 peak-sun hours ≈ 60 Wh/day; 50 Wh battery carries ~38 h of darkness

Four separate receivers would have doubled this budget. One receiver behind a switch adds nothing measurable, which is why the switch was chosen.

EXPECTED SERIAL OUTPUT

Detection line on USB at 115200, with the v2 keys appended to the existing contract:


```
{"type":"analog_fm","receiver":"rx5808","hw":"v2","mac":"AF:00:16:6C:52:03",
 "freq_mhz":5742,"band":"R","ch":3,"rssi_raw":903,"rssi":903,
 "rssi_mv":683,"rssi_dbm":-66.2,"rssi_min":880,"rssi_max":930,"rssi_n":20,
 "sectors":[-62.1,-68.4,-84.0,-79.2],"sector":0,"bearing_deg":41,"bearing_sigma_deg":14,
 "freq_peak":5742,"video":"NTSC","sync_hz":15736,"field_hz":60,"sync_q":96,"fp":"NTSC/15736/5742",
 "basic_id":"5.8G-R3-5742MHz","node_id":"RX01","seq":42}
```


Heartbeat every 60 s on USB, 120 s on the mesh:


```
{"heartbeat":true,"node_id":"RX01","receiver":"rx5808","hw":"v2","scanning":true,
 "channels":40,"sectors":4,"heading":0,"threshold":600,"threshold_dbm":-94.5,
 "video_seen":7,"temp_c":41.2,"uptime_s":3600,"seq":42}
```


> The v2 keys (`sectors` through `fp`) are additive and the mapper on `main` ignores them safely today; teaching its solver to use bearings and the fingerprint is the next `main` change. The mesh copy carries `rssi_dbm`, the bearing pair and `fp`, and is held under 200 bytes by dropping the fingerprint, then the bearing, rather than truncating.

RSSI CALIBRATION RECORD

Feed a known level through a step attenuator on a tuned channel, read `rssi_mv` off the serial monitor, and fill this in. Two points about 50 dB apart set `RSSI_CAL_MV_LO/HI` and `RSSI_CAL_DBM_LO/HI` in `rx5808.h`.

INPUT LEVEL (DBM)	−90	−80	−70	−60	−50	−40	−30
Measured `rssi_mv`							
Measured `rssi_raw`							

Noise floor with no transmitter: `rssi_raw` ______ `rssi_mv` ______ → set `RSSI_THRESHOLD` = floor + 200 = ______

4 · Full bill of materials

Single-unit planning prices, quantities per station. Total ≈ $190 all-in; ≈ $95 electronics only. The v1 station was ≈ $45 without power.

RF FRONT END

#	PART	QTY	NOTES	✓	≈ $
1	**RX5808** receiver, SPI-modded	1	Same module as v1. Remove the SPI-disable resistor if the batch ships with it; keep the u.FL pad for the switch feed.	☐	8
2	**5.8 GHz patch, 8 dBi, ~70°, SMA**	4	Linear is polarisation-agnostic (−3 dB vs an RHCP VTX); RHCP gains 3 dB on the common case, loses ~20 dB on LHCP. Four identical units from one batch.	☐	32
3	**SP4T RF switch**, 0.1–6 GHz, 3.3 V control	1	SKY13322-375LF or PE42442. ~0.5 dB loss at 5.8 GHz, sub-µs switching. Verify control line count and truth table. Prototype on the vendor eval board (~$40).	☐	3
4	u.FL–SMA pigtails, **equal length ≤ 15 cm**	5	Four patches plus switch common. Matched length keeps sector losses matched — this is what the bearing depends on.	☐	10
5	SMA bulkhead feedthroughs	4	One per enclosure face.	☐	8
6	5.8 GHz band-pass filter *(optional)*	1	Between switch and receiver if a nearby 2.4 GHz access point overloads the front end.	☐	5

VIDEO FINGERPRINT

#	PART	QTY	NOTES	✓	≈ $
7	**TI LMH1980** sync separator (SOT-23-6)	1	3.3 V supply and 3.3 V outputs straight to the XIAO; auto-detects SD/HD. Preferred.	☐	3
7b	*alt:* LM1881N (DIP-8)	1	5 V part: needs the 5 V rail plus a 2:1 divider or a 74LVC1G17 on each output. Fine for a breadboard.	☐	2
8	0.1 µF coupling cap, 75 Ω termination	1 ea	Per datasheet; the LM1881 also needs 680 kΩ RSET + 0.1 µF.	☐	1

MCU, MESH, POWER, MECHANICAL

#	PART	QTY	NOTES	✓	≈ $
9	**Seeed XIAO ESP32-S3** or **ESP32-C5**	1	Firmware picks the pinout at compile time.	☐	9
10	**Heltec WiFi LoRa 32 V3** + antenna	1	Meshtastic, serial module TEXTMSG 115200, node name = `NODE_ID`.	☐	20
11	**Solar panel 20 W, 18 V**	1	South-facing, 30–45° tilt. 10 W covers summer only.	☐	25
12	**MPPT charger, 5 V / 1 A out**	1	DFRobot DFR0559 or Waveshare Solar Power Manager.	☐	15
13	**18650 Li-ion 3500 mAh**, 1S4P holder	4	≈ 50 Wh. Protected cells or a BMS board.	☐	24
14	1000 µF low-ESR cap at the Heltec	1	LoRa TX bursts.	☐	1
15	**IP65 enclosure** ≈ 200×150×100 mm	1	Patches on the four vertical faces at 0/90/180/270°, each tilted 10–15° up.	☐	15
16	Cable gland, desiccant	1		☐	3
17	Pole bracket + four patch wedges	1 set	**Face N to true north and record the heading** — a mis-oriented box rotates every bearing it reports.	☐	10
18	2 kΩ ×3, 100 nF ×4, 10 µF	—	Switch control series resistors and supply decoupling.	☐	1

5 · Bench procedure

Work top to bottom. Do not move to the next stage until the current one gives the stated result — a bearing built on an uncharacterised switch is not a bearing.

STAGE 1 — RF SWITCH

- ☐ Eval board powered, one patch on RF1, RX5808 on the common port, XIAO GPIOs D1/D2/D3 to the control lines through 2 kΩ.
- ☐ Drive each control combination and record which RF port goes through. **Write the truth table here:**
  `V1V2V3 = 000 → RF__   001 → RF__   010 → RF__   011 → RF__`
- ☐ With a VTX on a known channel, scope the RSSI line while toggling sectors. **Expected:** settled within 1 ms. Measured: ________ µs
- ☐ Compare RSSI with the switch in circuit against the patch wired direct. **Expected:** ≈ 0.5 dB loss. Measured: ________ dB

STAGE 2 — SYNC SEPARATOR

- ☐ RX5808 VIDEO → 0.1 µF → separator input, 75 Ω terminated. Scope CSYNC with a real VTX on the tuned channel. **Expected:** clean train at 15.6–15.7 kHz. Measured: ________ Hz
- ☐ Scope VSYNC. **Expected:** 50 Hz (PAL) or 59.94 Hz (NTSC). Measured: ________ Hz
- ☐ Tune to a channel carrying only Wi-Fi (5745–5825 MHz band). **Expected:** no stable sync train. Confirmed: yes / no
- ☐ Confirm output logic levels reach the XIAO's input threshold (LM1881 at 5 V needs the divider).

STAGE 3 — RSSI CALIBRATION

- ☐ Fill the calibration table on sheet 4 using a step attenuator, or a VTX at two known open-field distances.
- ☐ Enter the two points in `rx5808.h` as `RSSI_CAL_MV_LO/DBM_LO` and `RSSI_CAL_MV_HI/DBM_HI`, then reflash.
- ☐ Set `RSSI_CAL_OFFSET_DB` per station so a fleet agrees on the same source. Value used: ________ dB
- ☐ Confirm the heartbeat now reports a sane `threshold_dbm` (expect roughly −90 to −95). Reported: ________ dBm

STAGE 4 — PATTERN CALIBRATION

- ☐ Mount all four patches on the box, VTX at 30 m on a known bearing, clear line of sight.
- ☐ Rotate the box in 15° steps through a full turn, logging all four `sectors` values at each step.
- ☐ Derive κ in degrees per dB from the neighbour difference. Value: ________ °/dB
- ☐ Save the sweep to `docs/` as the station's bearing calibration.

STAGE 5 — FIELD ACCEPTANCE

- ☐ Two v2 stations 300–500 m apart, both positions set in the mapper, both named to match their Meshtastic nodes.
- ☐ Walk a VTX between them on ten marked points. **Pass:** the 90 % circle contains the true position on at least 9 of 10. Score: ____ / 10
- ☐ **Pass:** `video` reads NTSC or PAL on every report, never `none`, for a live VTX.
- ☐ Leave both stations on solar for 72 h. **Pass:** no brownout, heartbeat gap never exceeds 5 min.

Built by		**Date**	
Verified by		**Date**	
Station ID		**True-north heading of face N**	